IGCSE 0607 Practice Questions — E1.11

Rates

Total Questions: 9 — **Total Marks:** 46 Marks

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1 A train departs at 08 : 15 and arrives at its destination at 09 : 45.
The train travels a distance of 120 km.
Calculate the average speed of the train.

..... km/h **[2]**

2 A car travels 120 km in 1 hour and 30 minutes.
Calculate the average speed of the car in km/h.

..... km/h **[3]**

3 A train travels a distance of 385 km.
The journey takes 4 hours 25 minutes.
Calculate the average speed of the train in km/h.
Give your answer correct to 1 decimal place.

..... km/h **[3]**

4 Water flows from a pipe into a rectangular tank at a rate of 45 litres per minute.

(a) Calculate the volume of water, in cubic centimetres (cm^3), that flows into the tank in 1 second.

..... cm^3 **[2]**

(b) The tank has a base measuring 1.5 m by 80 cm. Calculate the rate at which the water level is rising in the tank, giving your answer in cm/s.

..... cm/s **[3]**

5 A car uses fuel at an average rate of 6.4 litres per 100 km.

(a) Calculate the amount of fuel used for a journey of 350 km.

..... litres **[2]**

(b) The cost of fuel is 1.35 euros per litre.
The exchange rate is \$1 = 0.85 euros. Calculate the cost of the fuel for this journey in dollars (\$).
Give your answer correct to the nearest cent.

\$ ..... **[3]**

6 Water flows from a pipe into a rectangular tank at a rate of 2.5 litres per second.
The tank has a length of 1.8 m, a width of 1.5 m, and a height of 1.2 m.
The tank is initially empty.

(a) Calculate the capacity of the tank in litres.

..... litres **[2]**

(b) Work out how long it takes to fill the tank completely.
Give your answer in minutes.

..... minutes **[2]**

(c) Another pipe empties the full tank at a rate of x litres per minute.
It takes 45 minutes to empty the tank completely.
Calculate the value of x.

$x =$ ..... **[2]**

7 A car travels from City A to City B, a distance of d km, at an average speed of v km/h.
The car then travels from City B to City C, a distance of $2d$ km, at an average speed of $(v + 15)$ km/h.
The total time taken for the entire journey from City A to City C is T hours.

(a) Show that the total time T can be written as

$$T = \frac{3dv + 15d}{v(v + 15)}$$

[3]

(b) Given that $d = 120$ and $T = 4.5$,

Form an equation in v and show that it simplifies to

$$3v^2 - 195v - 1200 = 0$$

[4]

8 A cyclist travels from Town A to Town B, a distance of 48 km.
The cyclist travels the first 30 km at an average speed of x km/h. They travel the remaining distance at an average speed of $(x - 4)$ km/h. The total time taken for the journey is 2 hours and 30 minutes.

(a) Show that $5x^2 - 116x + 240 = 0$.

[4]

(b) Solve the equation $5x^2 - 116x + 240 = 0$ to find the possible values of x.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... **[3]**

9 A man runs 15 km at an average speed of x km/h.
He then walks 12 km at an average speed of $(x - 4)$ km/h.
The total time taken for the journey is 3.5 hours.

(a) Show that x satisfies the equation

$$7x^2 - 82x + 120 = 0$$

[3]

(b) **(i)** Solve the equation

$$7x^2 - 82x + 120 = 0$$

$x =$ ..... , or $x =$ ..... **[3]**

(ii) Find the time taken for the man to run the 15 km.

..... hours **[2]**